

ALEXANDER REED

Senior Machine Learning Engineer (NLP & Generative AI)

Professional Summary

Senior ML Engineer with 5+ years of experience building NLP, LLM, and Retrieval-Augmented Generation (RAG) systems at scale. Experienced in PyTorch, JAX, LangChain, LangGraph, Hugging Face, vector databases, distributed training, MLOps, and cloud-native AI deployments. Led development of enterprise AI assistants, document intelligence platforms, and production-grade LLM applications.

Technical Skills

Python, PyTorch, JAX, TensorFlow, NLP, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Prompt Engineering, LangChain, LangGraph, Hugging Face Transformers, FAISS, ChromaDB, Pinecone, Vector Search, Kubernetes, Docker, AWS, Azure, FastAPI, MLflow, Airflow, CI/CD, GitHub Actions.

Professional Experience

Senior ML Engineer – NLP & GenAI (2023–Present)

- Designed enterprise RAG systems processing millions of documents.
- Built LangGraph-based multi-agent workflows for customer support automation.
- Fine-tuned transformer models using PyTorch and JAX.
- Reduced inference latency by 35% through model optimization and caching.

Machine Learning Engineer (2021–2023)

- Developed NLP pipelines for text classification, NER, semantic search, and summarization.
- Built production APIs serving LLM-powered applications using FastAPI and Kubernetes.
- Implemented vector search with FAISS and Pinecone.

Associate ML Engineer (2020–2021)

- Created document intelligence solutions using transformer architectures.
- Automated ML training and deployment workflows using MLOps practices.

Selected Projects

- Enterprise Knowledge Assistant using LangChain, LangGraph, Pinecone, and GPT-based models.
- Legal Document Intelligence Platform with RAG and semantic retrieval.
- Multi-Agent Research Assistant for automated information synthesis.
- Real-Time Resume Screening System powered by embeddings and vector search.

Education

Master of Science in Artificial Intelligence
Bachelor of Science in Computer Science